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open type without heart and blood vessels. Sexes are separate. Reproduction is sexual and fertilization is external. Development is indirect with free swimming bilaterally symmetrical larval forms. Some echinoderms exhibit autotomy with remarkable powers of regeneration. (Figure 2.17)

Examples: *Asterias* (Starfish or sea star), *Echinus* (Sea-urchin), *Antedon* (Sea-lily), *Cucumaria* (Sea-cucumber), *Ophiura* (Brittle star)

2.3.10 Phylum: Hemichordata

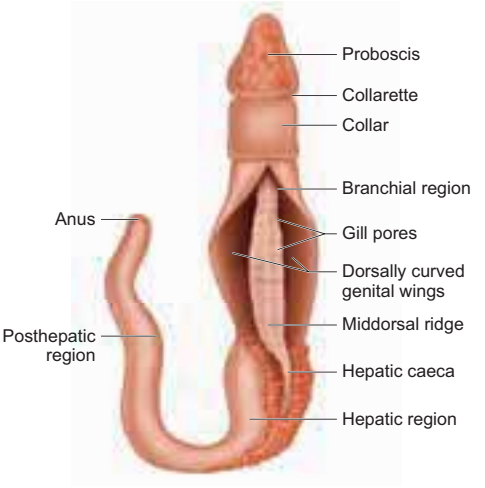
(G.*hemi* –half; *chorde*-string)

Hemichordates were earlier treated as a subphylum of Chordata (or Prochordata). They are now regarded to be an independent phylum of invertebrates, close to Echinodermata. The animals of this group possess the characters of invertebrates as well as chordates.

This phylum consists of a small group of worm-like, soft marine animals, mostly tubiculous and commonly called the 'acorn worms' or 'tongue worms'. They are bilaterally symmetrical, triploblastic and coelomate animals with organ system level of organisation. Their body is cylindrical and is divided into three regions, the anterior proboscis, a short collar and a long trunk. Most hemichordates are ciliary feeders. Their circulatory system is simple and open or lacune type with a dorsal heart. Respiration is through paired gill slits opening into the pharynx. Excretion is by a single proboscis gland or glomerulus situated in the proboscis. Nervous system is primitive. Sexes are separate and exhibit sexual mode of reproduction; Fertilization is external. Development is indirect with a free swimming tornaria larva.



Balangoglossus



Diagrammatic representation of Balanoglossus

Figure 2.18 Example of Hemichordata

Examples: *Balanoglossus*, *Saccoglossus*, *Ptychodera flava* (Indian Hemichordate found in Kurusadai islands in Tamilnadu) (Figure 2.18).

2.4 Phylum: Chordata

(G. *Chorda –string*)

Chordata is the largest phylum with most familiar group of animals, such as fishes, amphibians, reptiles, birds and mammals and less known forms such as lancelets (Amphioxus) and tunicates (Ascidian). All chordates possess three fundamental distinct features at some stage of their life cycle (Figure 2.19), they are: